

JAMES RUSE AGRICULTURAL HIGH SCHOOL

3/4 Unit Mathematics

Year 12 Term 1 Assessment 2000

TIME ALLOWED : 85 Minutes

- Start each question on a new page
- All questions are of equal value
- Each question is to be handed in separately

QUESTION 1:(9 marks) Start this question on a new page

- a) Solve for x : $\log_2 x + 3\log_2 4 = \log_2 128$
- b) Differentiate with respect to x : $y = \cos^5 2x$
- c) Find the equation of the normal to the curve $y = e^{3x}$ at the point where $x = \frac{1}{3}$.

QUESTION 2:(9 marks) Start this question on a new page

- a) Find the following indefinite integral: $\int \frac{x+1}{x^2+2x-9} dx$
- b) Prove by mathematical induction that $2^{3n} - 3^n$ is always divisible by 5.
- c) Find the area bounded by $y = \sin x$, $y = \tan x$ and the line $x = \frac{\pi}{4}$.

QUESTION 3:(9 marks) Start this question on a new page

- a) Evaluate the following definite integral using the substitution given:

$$\int_0^{\frac{\pi}{2}} \cos x \sqrt{(\sin x)^3} dx ; u = \sin x$$

- b) Differentiate with respect to x : $y = \ln \left(\frac{\sqrt{x+1}}{2x-1} \right)$
- c) The value of a car when new is \$45 000. If it depreciates at the rate of 18% of its value at the beginning of each year, find its value after 8 years (answer to the nearest dollar)

QUESTION 4:(9 marks) Start this question on a new page

- (a) Find the volume of the solid of revolution when the area bounded by the curve $y = \cos 3x$, the x -axis and $x=0$ and $x=\frac{\pi}{6}$ is rotated about the x -axis.
- (b) (i) Express $\sqrt{3} \cos x - \sin x$ in the form $A \cos(x + \alpha)$ where $A > 0$ and α is acute.
- (ii) Hence, solve the equation $\sqrt{3} \cos x - \sin x = -1$ for $0 \leq x \leq 2\pi$
- (iii) For $y = \sqrt{3} \cos x - \sin x$ find values of x in the domain $0 \leq x \leq 2\pi$ for which this function is a maximum.

QUESTION 5:(9 marks) Start this question on a new page

- (a) Sketch the graph of the function $y = \log_e(x-2)$.
- (b) Rotate about the y -axis the region bounded by the curve $y = \log_e(x-2)$, $y=0$ and $y=h$ to create a bowl. Find the exact volume of the bowl.
- (c) The bowl is placed with its axis vertical and water is poured in. If water is poured into the bowl at a rate of 50 cm^3 per second, find the rate at which the water level is rising when the depth of water is 1.5 cm (answer to 3 decimal places).

QUESTION 6:(9 marks) Start this question on a new page

- (a) (i) Show that $\frac{d}{dx} \tan^3 x = 3 \sec^4 x - 3 \sec^2 x$.
- (ii) Hence, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x dx$.
- (b) (i) Find the difference between the simple interest and compound interest on \$5000 invested at 6% p.a. for 4 years (answer to the nearest cent).
- (ii) What is the equivalent simple interest rate to earn this compound interest on the same principle over the same time?

QUESTION 7:(9 marks) Start this question on a new page

- (a) On January 1st 2000, Sue Bright invested \$1000 in a superannuation scheme. On January 1st of each of the subsequent 14 years, she will make further investments, increasing them by 5% each year to account for inflation. The scheme pays 10% per annum interest, calculated annually, and she will withdraw her funds when the scheme reaches maturity on January 1st, 2015.

Find:

- the value of her first investment when it is withdrawn.
 - the value of her last investment when it is withdrawn.
 - to the nearest dollar, the amount she will withdraw on January 1st, 2015.
- (b) A wooden beam is cut from a solid log so that the cross section of the log is as follows:

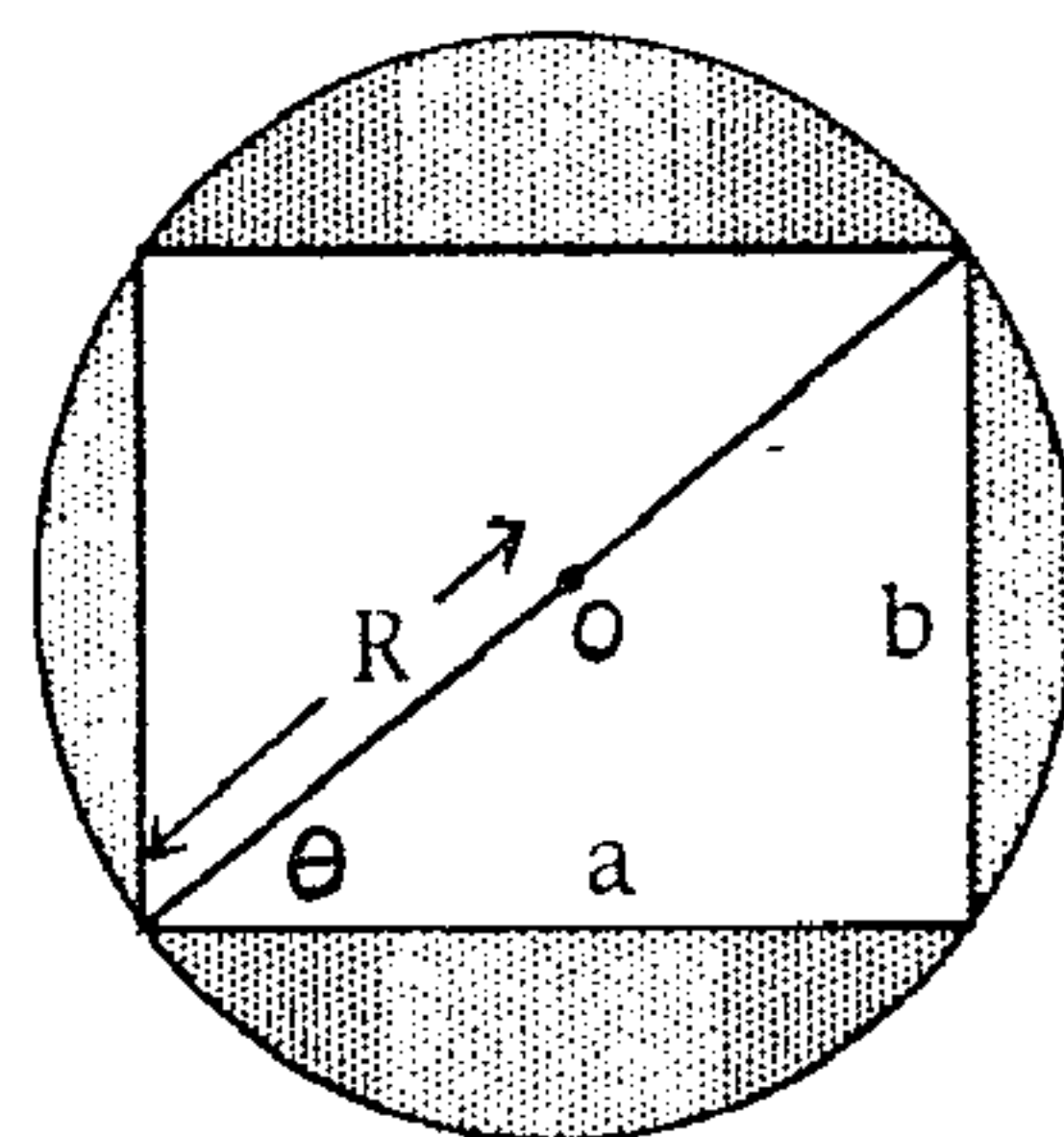


Diagram not to scale

The wooden rectangular beam of ^{width} ~~length~~ a cm and height b cm is cut from a circular log of fixed radius R cm. The strength S , of a rectangular beam is given by the formula $S = ka^2b$ where k is a constant and $k > 0$.

- Show that the strength of this beam, which can be cut from the circular log has equation $S = 8R^3k \sin \theta \cos^2 \theta$.
- Find the value of θ , in radians to 3 decimal places, that would maximise the strength of the beam.

THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Qu1

a) $\log_2 x + \log_2 4^3 = \log_2 128$
 $\log_2(64x) = \log_2 128$
 $x = 2$ (3)

b) $y = \cos^5 2x$
 $\frac{dy}{dx} = 5 \cos^4 2x \cdot -2 \sin 2x$
 $= -10 \sin 2x \cdot \cos^4 2x$ (3)

c) $y = e^{3x}$
 $\frac{dy}{dx} = 3e^{3x}$
 at $x = \frac{1}{3}$: $m = 3e$
 $\therefore \frac{1}{m} = -\frac{1}{3e}$
 at $x = \frac{1}{3}$: $y = e$
 $y - e = -\frac{1}{3e}(x - \frac{1}{3})$
 OR
 $y = -\frac{1}{3e}x + \frac{1}{9e} + e$ (3)

Qu2

a) $\int \frac{x+1}{x^2+2x-9} dx$
 $= \frac{1}{2} \ln(x^2+2x-9) + C$ (2)

b) Show true for $n=1$
 $2^3 - 3 = 5$ which is $\div$ by 5
 Assume true for $n=k$
 i.e. $2^{3k} - 3^k = 5M$ ($M \in \mathbb{J}$)
 Show true for $n=k+1$
 i.e. $2^{3(k+1)} - 3^{k+1}$ is divisible by 5
 $2^{3k+3} - 3^{k+1} = 2^{3k} \cdot 2^3 - 3^k \cdot 3$
 $= 8(5M + 3^k) - 3 \cdot 3^k$
 (by assumption)
 $= 40M + 5 \cdot 3^k$
 $= 5(8M + 3^k)$
 which is divisible by 5
 as $8M + 3^k \in \mathbb{J}$

As the result is true for $n=1$ and
 $n=k+1$ assuming its true for $n=k$
 then it is true for $n=2, 3, \dots$
 and all positive integer values of n (4)

c) $A = \int_0^{\frac{\pi}{4}} (\tan x - \sin x) dx$
 $= \int_0^{\frac{\pi}{4}} \left(\frac{\sin x}{\cos x} - \sin x \right) dx$
 $= \left[-\ln \cos x + \cos x \right]_0^{\frac{\pi}{4}}$
 $= \left(-\ln \cos \frac{\pi}{4} + \cos \frac{\pi}{4} \right) -$
 $\left(-\ln \cos 0 + \cos 0 \right)$
 $= \ln \sqrt{2} + \frac{\sqrt{2}}{2} - 1$ (3)

Qu3

a) $\int_0^{\frac{\pi}{2}} \cos x \cdot \sqrt{(\sin x)^3} dx$
 $u = \sin x$
 $du = \cos x dx$
 $x = \frac{\pi}{2}$ $u = 1$
 $x = 0$ $u = 0$
 $\therefore \int_0^1 u^{\frac{3}{2}} du$
 $= \left[\frac{2}{5} u^{\frac{5}{2}} \right]_0^1$
 $= \frac{2}{5}$ (3)

b) $y = \ln \left(\frac{\sqrt{x+1}}{2x-1} \right)$
 $= \frac{1}{2} \ln(x+1) - \ln(2x-1)$
 $\frac{dy}{dx} = \frac{1}{2} \cdot \frac{1}{x+1} - \frac{2}{2x-1}$
 $= \frac{-2x-5}{2(x+1)(2x-1)}$ (3)

(2)

c) $I = 45000 \left(1 - \frac{18}{100} \right)^8$
 $= \$9198.63$
 $= \$9199$ (nearest \$) (3)

Qu4

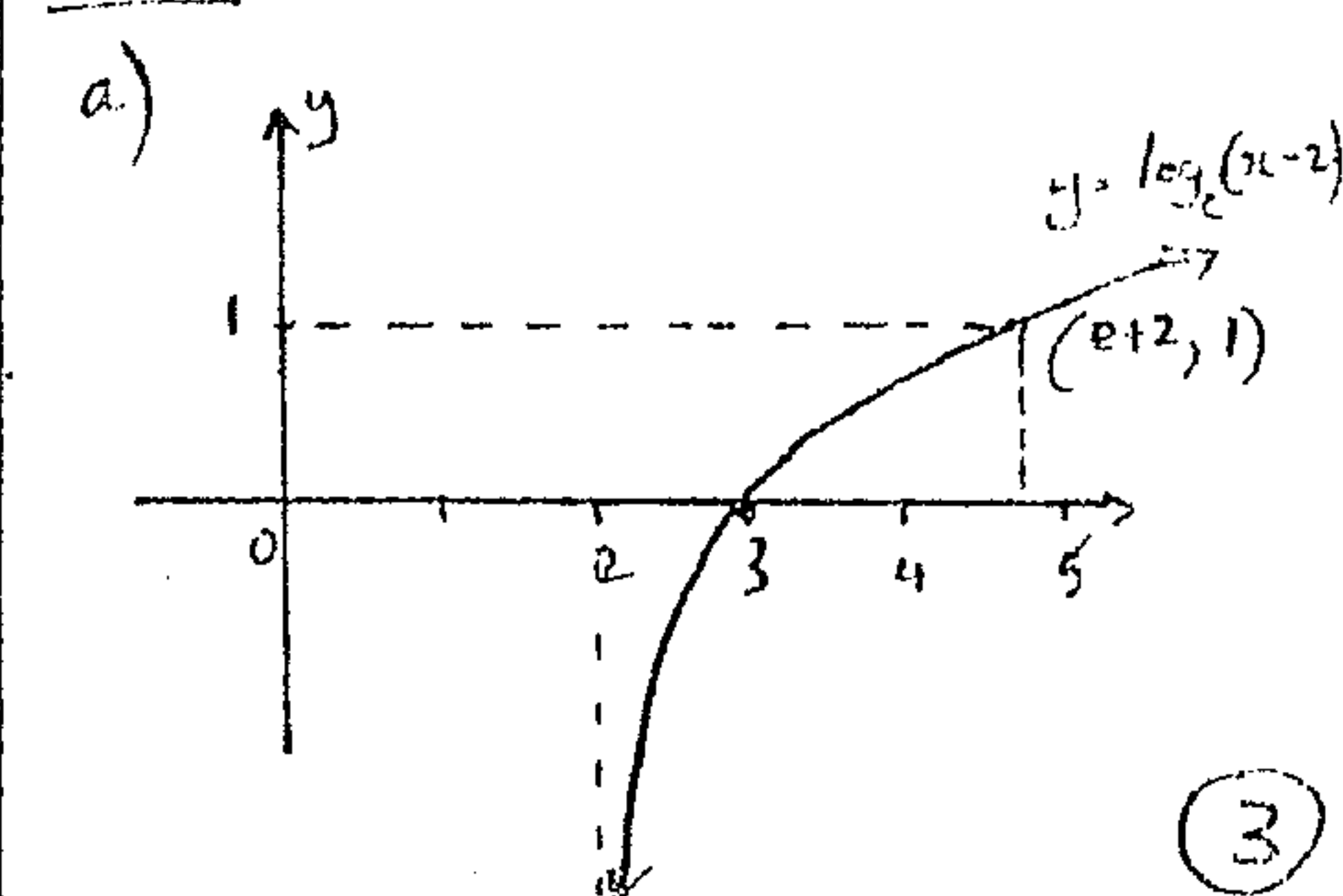
a) $V = \pi \int_0^{\frac{\pi}{6}} \cos^2 3x dx$
 $= \frac{\pi}{2} \int_0^{\frac{\pi}{6}} (\cos 6x + 1) dx$
 $= \frac{\pi}{2} \left[\frac{1}{6} \sin 6x + x \right]_0^{\frac{\pi}{6}}$
 $= \frac{\pi}{2} \left[\left(\frac{1}{6} \sin \pi + \frac{\pi}{6} \right) - 0 \right]$
 $= \frac{\pi^2}{12}$ (3)

b) (i) $\sqrt{3} \cos x - \sin x = A \cos(x+\alpha)$
 $A = \sqrt{3+1} = 2$
 $\alpha = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}$
 $\therefore \sqrt{3} \cos x - \sin x = 2 \cos \left(x + \frac{\pi}{6} \right)$ (2)

(ii) $\sqrt{3} \cos x - \sin x = -1$
 $2 \cos \left(x + \frac{\pi}{6} \right) = -1$
 $\cos \left(x + \frac{\pi}{6} \right) = -\frac{1}{2}$
 For acute $x + \frac{\pi}{6}$, $x + \frac{\pi}{6} = \frac{\pi}{3}$
 $\cos \left(x + \frac{\pi}{6} \right)$ is -ve in 2nd & 3rd quad.
 $\therefore x + \frac{\pi}{6} = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$
 $x = \pi - \frac{\pi}{3} - \frac{\pi}{6}, \pi + \frac{\pi}{3} - \frac{\pi}{6}$
 $x = \frac{\pi}{2}, \frac{3\pi}{2}$ (2)

(iii) Max when $2 \cos \left(x + \frac{\pi}{6} \right) = 2$
 $\cos \left(x + \frac{\pi}{6} \right) = 1$
 $x + \frac{\pi}{6} = 2\pi$
 $x = 11\pi$ (2)

Qu5



b) $y = \log_e(x-2)$
 $e^y = x-2$
 $\therefore x = e^y + 2$
 $V = \pi \int_0^h (e^y + 2)^2 dy$
 $= \pi \int_0^h (e^{2y} + 4e^y + 4) dy$
 $= \pi \left[\frac{1}{2} e^{2y} + 4e^y + 4y \right]_0^h$
 $= \pi \left(\frac{1}{2} e^{2h} + 4e^h + 4h \right) - \pi \left(\frac{1}{2} + 4 \right)$
 $= \pi \left(\frac{1}{2} e^{2h} + 4e^h + 4h - \frac{9}{2} \right)$

c) $\frac{dV}{dt} = 50 \text{ cm}^3/\text{sec}$ (3)
 Need $\frac{dh}{dt}$ when $h = 1.5 \text{ cm}$.
 $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$
 $50 = \pi (e^{2h} + 4e^h + 4) \cdot \frac{dh}{dt}$
 at $h = 1.5$: $\frac{dh}{dt} = \frac{50}{\pi (e^3 + 4e^{1.5} + 4)}$
 $= 0.379 \text{ cm/sec}$
 (to 3dp)

$\therefore$ Water is rising at a rate of
 0.379 cm/sec

Qu 6

a) (i) $\frac{d}{dx}(\tan^3 x)$

$$= 3 \tan^2 x \cdot \sec^2 x$$

$$= 3 (\sec^2 x - 1) \cdot \sec^2 x$$

$$= 3 \sec^4 x - 3 \sec^2 x$$

(2)

(ii) $\frac{d}{dx}(\tan^3 x) = 3 \sec^4 x - 3 \sec^2 x$

$$\frac{d}{dx}(\tan^3 x) + 3 \sec^2 x = 3 \sec^4 x$$

$$\therefore \int 3 \sec^4 x dx = \left[\tan^3 x + 3 \tan x \right]_0^{\pi/4}$$

$$\therefore \int \sec^4 x dx = \frac{1}{3} \left[\tan^3 x + 3 \tan x \right]_0^{\pi/4}$$

$$= \frac{4}{3}$$

(3)

b) (i) $SI = 5000 \times \frac{6}{100} \times 4 = \1200

$$CI = 5000(1.06)^4 - 5000$$

$$= \$1312.38$$

$$\therefore \text{Difference} = 1312.38 - 1200$$

$$= \$112.38$$

(2)

(ii) $1312.38 = 5000 \times \frac{R}{100} \times 4$

$$R = \frac{1312.38 \times 100}{5000 \times 4}$$

$$R = 6.56$$

$$\therefore \text{Equivalent simple interest rate is } \underline{6.56\%}$$

(2)

Qu 7 a)

(i) $\frac{1000}{1000} (1.1)^{15}$ 15

(ii) $\frac{1000}{1000} (1.05)^{14} (1.1)$ 14

(iii) $\frac{1000}{1000} (1.05)^{13} (1.1)^2$ 13

(iv) $\frac{1000}{1000} (1.05)^{12} (1.1)^3$ 12

(v) $\frac{1000}{1000} (1.05)^{11} (1.1)^4$ 11

(vi) $\frac{1000}{1000} (1.05)^{10} (1.1)^5$ 10

(vii) $\frac{1000}{1000} (1.05)^9 (1.1)^6$ 9

(viii) $\frac{1000}{1000} (1.05)^8 (1.1)^7$ 8

(ix) $\frac{1000}{1000} (1.05)^7 (1.1)^8$ 7

(x) $\frac{1000}{1000} (1.05)^6 (1.1)^9$ 6

(xi) $\frac{1000}{1000} (1.05)^5 (1.1)^{10}$ 5

(xii) $\frac{1000}{1000} (1.05)^4 (1.1)^{11}$ 4

(xiii) $\frac{1000}{1000} (1.05)^3 (1.1)^{12}$ 3

(xiv) $\frac{1000}{1000} (1.05)^2 (1.1)^{13}$ 2

(xv) $\frac{1000}{1000} (1.05)^1 (1.1)^{14}$ 1

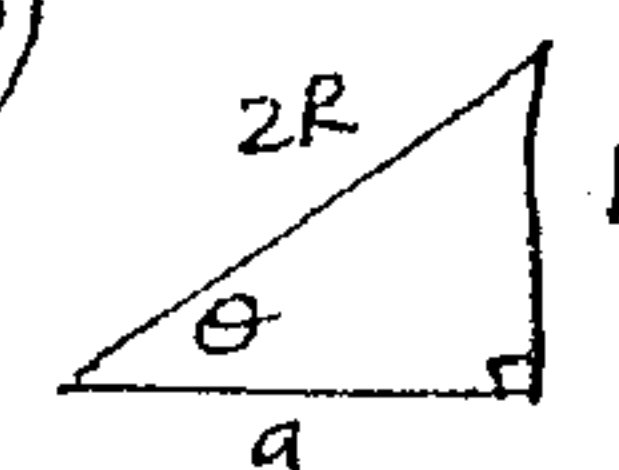
(xvi) $\frac{1000}{1000} (1.05)^0 (1.1)^{15}$ 0

AP where $a = 1.1^{15}$ $r = \frac{1.05}{1.1}$ $n = 15$

$$= 1000 \left[\frac{1.1^{15} \left(\frac{1.05}{1.1} - 1 \right)}{\frac{1.05}{1.1} - 1} \right] = \$46163$$

(2)

b)



$$\cos \theta = \frac{a}{2R}$$

$$\sin \theta = \frac{b}{2R}$$

$$a = 2R \cos \theta$$

$$b = 2R \sin \theta$$

(i) $S = k a^2 b$

$$= k (2R \cos \theta)^2 (2R \sin \theta)$$

$$S = 8k R^3 \cos^2 \theta \sin \theta$$

$$0 < \theta < \frac{\pi}{2}$$

(2)

(ii) $S = 8R^3 k \sin \theta \cos^2 \theta$

$$\frac{dS}{d\theta} = 8R^3 k [\sin \theta \cdot 2 \cos \theta \cdot -\sin \theta + \cos^2 \theta \cdot \cos \theta]$$

$$= 8R^3 k [-2 \sin^2 \theta \cos \theta + \cos^3 \theta]$$

(3)

For max/min: $\frac{dS}{d\theta} = 0$

$$8R^3 k [\cos^3 \theta - 2 \sin^2 \theta \cos \theta] = 0$$

$$\cos \theta (\cos^2 \theta - 2 \sin^2 \theta) = 0$$

$$\cos \theta = 0$$

$$\theta = 90$$

$$\text{but } \theta \neq 90$$

$$\cos^2 \theta - 2 \sin^2 \theta = 0$$

$$1 - \sin^2 \theta - 2 \sin^2 \theta = 0$$

$$1 - 3 \sin^2 \theta = 0$$

$$\sin^2 \theta = \frac{1}{3}$$

$$\sin \theta = \pm \frac{1}{\sqrt{3}}$$

$$\theta \text{ can only be acute}$$

$$\therefore \theta = 0.615$$

Check $\theta = 0.615$ is a max:

$$\frac{d^2 S}{d\theta^2} = 8R^3 k [-2(\sin^2 \theta \cdot -\sin \theta + \cos \theta \cdot 2 \sin \theta \cos \theta) + 3 \cos^2 \theta \cdot -\sin \theta]$$

$$= 8R^3 k [2 \sin^3 \theta - 4 \sin \theta \cos^2 \theta - 3 \sin \theta \cos^2 \theta]$$

$$= 8R^3 k [2 \sin^3 \theta - 7 \sin \theta \cos^2 \theta]$$

at $\theta = 0.615$, $\frac{d^2 S}{d\theta^2} = 8R^3 k (-2.310)$

$$< 0 \text{ as } R^3 > 0 \text{ and } k > 0$$

$$\Rightarrow \text{concave down}$$

$$\Rightarrow \text{local max at } \theta = 0.615$$

Since this is a continuous function and there is only 1 maximum value, then $\theta = 0.615$ is the absolute maximum.